How many coin tosses would you need until you get n Heads or m Tails?

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–Joint work with Svante Janson and Doron Zeilberger–

Background and History

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The mail correspondence between Fermat and Pascal laid the foundation for what is now known as probability theory.

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Convention: For the rest of the talk, p represents the probability of getting Heads and $q := 1 - p$ represents the probability of getting Tails.

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One of the (many) solutions provided by Fermat and Pascal was to split the prize with Player 1 receiving the above expression, and Player 2 receiving $1 - A(n,m,1)$.

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Remark: If we extend the above expression to infinity, then it has the following *closed-form*: $\left(\frac{px}{1-qx}\right)^n$.

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Remark: Fermat and Pascal were only interested in the probability of getting n Heads vs. getting m Tails.

Computations

Probability generating function (n Heads OR m Tails)

$X := X(n, m;\ p)$: The number of tosses until reaching (for the first time) either n Heads OR m Tails, where with probability p you get Heads.

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Remark: This becomes computationally slow. Using Wilf-Zeilberger algorithmic proof theory, we will obtain a third-order linear recurrence!

Results

Recurrence using Wilf-Zeilberger theory

Let $L(n, m;\ p)$ be the expectation of $X := X(n, m;\ p)$, which is the number of tosses until reaching (for the first time) either n Heads OR m Tails. Then,

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$$\begin{aligned} L(n, m;\ p) = & \frac{(pn + pm - 2p + 2n - 2)}{n-1} \cdot L\left(n-1, m\right) \\ & - \frac{(2pn + 2pm - 4p + n - 1)}{n-1} \cdot L\left(n-2, m\right) \\ & + \frac{p\left(m - 2 + n\right)}{n-1} \cdot L\left(n-3, m\right), \end{aligned}$$

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with initial conditions:

$$\begin{aligned} & L(1,1) = 1, L(1,2) = 2 - p, L(1,3) = p^2 - 3p + 3, L(2,1) = p + 1, \\ & L(2,2) = -2p^2 + 2p + 2, L(2,3) = 3p^3 - 7p^2 + 3p + 3 \\ & L(3,1) = p^2 + p + 1, L(3,2) = -3p^3 + 2p^2 + 2p + 2, \\ & L(3,3) = 6p^4 - 12p^3 + 3p^2 + 3p + 3. \end{aligned}$$

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	Expected coin tosses
$n = 1$	3.272000000
$n = 2$	7.491773440
$n = 3$	11.90103592
$n = 4$	16.40588424
$n = 5$	20.97105153
$n = 6$	25.57874312
$n = 7$	30.21849553
$n = 8$	34.88353902
$n = 9$	39.56919552
$n = 10$	44.27207236

Table: The probability of getting Heads is $p = 2/5$.

Explicit Expression (an Heads OR bn Tails)

Fix positive integers a, b. Let $p = \frac{a}{a+b}$ be the probability of getting Heads. If your goal is to reach either an Heads OR bn Tails, then the expected duration is given by

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One might be interested in the **asymptotics** of $L(an, bn, \frac{a}{a+b})$.

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$$L(an, bn, \ \tfrac{a}{a+b}) \sim (a+b)n \left(1 - \sqrt{\frac{a+b}{2ab\pi}} \cdot \frac{1}{\sqrt{n}}\right), \qquad n \to \infty.$$

Connection to other work

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Back to our language: You toss a coin until you get n Heads OR n Tails, where with probability p you get Heads.

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Let $\mathbb{E}_{n,p}$ be the expectation of (the number of Heads - the number of Tails) after $X(n, n;\ p)$ tosses i.e. once we reach either n Heads OR n Tails. Then,

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$$\mathbb{E}_{n,p} = n(p-q)\sum_{j=0}^{n-1} C_j (pq)^j,$$

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Their result is equivalent to our result in the special case of reaching n Heads OR n Tails.

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Let $X_1, X_2, \ldots$ be independent, identically distributed random variables with finite mean μ and let $S_n (n \geqslant 1)$, denote the partial sums. Suppose that τ is a stopping time. If $\mathbb{E}[\tau] < \infty$, then

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- Hence,

$$\mathbb{E}[S_\tau] = \mu \, \mathbb{E}[\tau] = (p - q) L(n, n;\ p)$$

Takeaway: total average profit= (average gain per toss)(average number of tosses)

Back to Volkov and Wiktorsson

From Wald's identity, $\mathbb{E}_{n,p} = \mathbb{E}\left[S_\tau\right] = (p-q)L(n,n;\ p)$.

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In the fair case, $p = \frac{1}{2}$, we can verify that with $a = b = 1$, we get

$$L(n,n;\ \tfrac{1}{2}) = 2n\left(1 - \frac{(2n)!}{n!^2}4^{-n}\right).$$

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If you would like to read the paper:

`https://marti310.github.io/research.html`

Thank you!

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Results (2)

Expected coin tosses figures for $n = m$

Using the recurrence for the expectation, we can compute the following with various p values.

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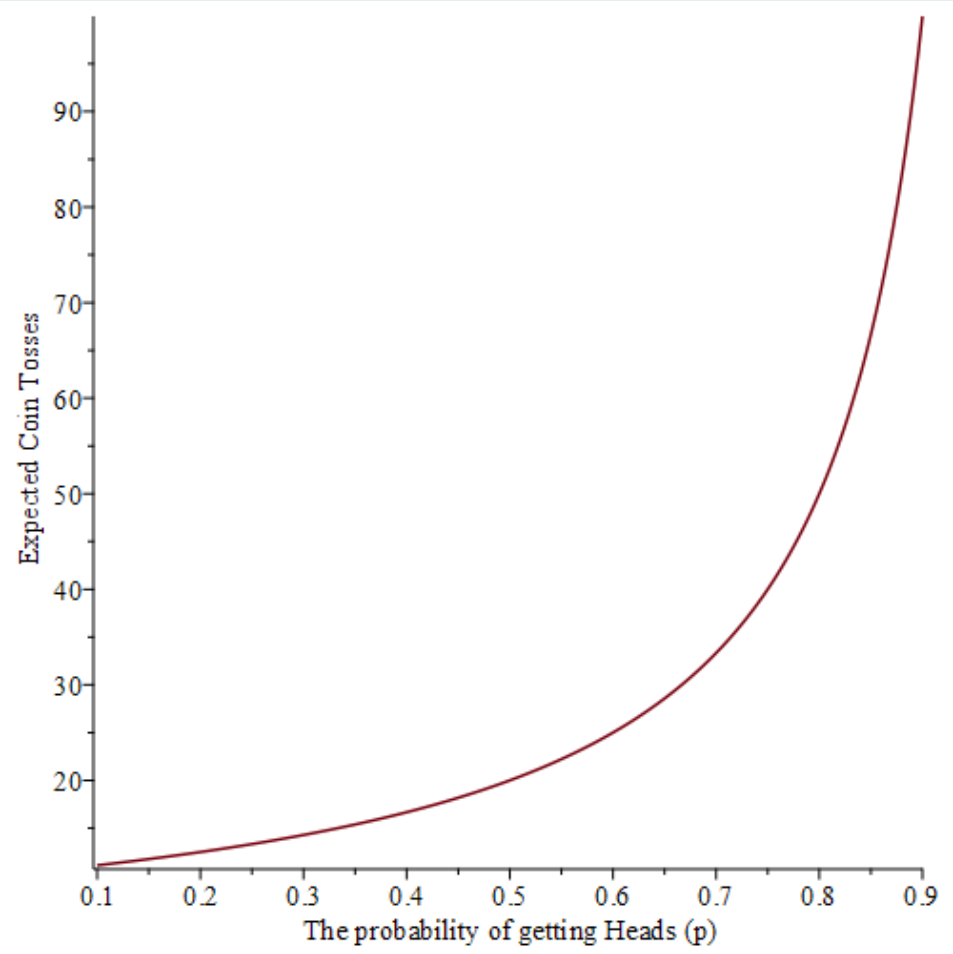
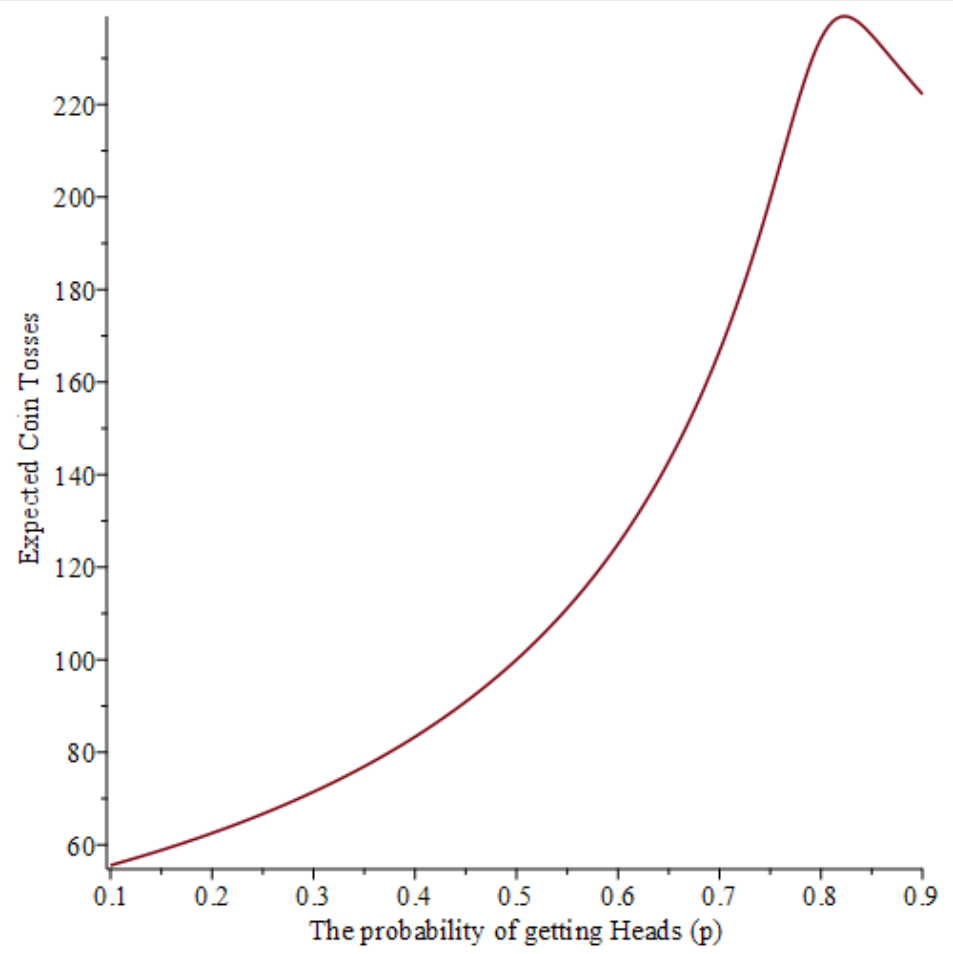
If your goal is to reach either 200 Heads OR 10 Tails,



Expected coin tosses figures for $n = m$

Using the recurrence for the expectation, we can compute the following with various p values.

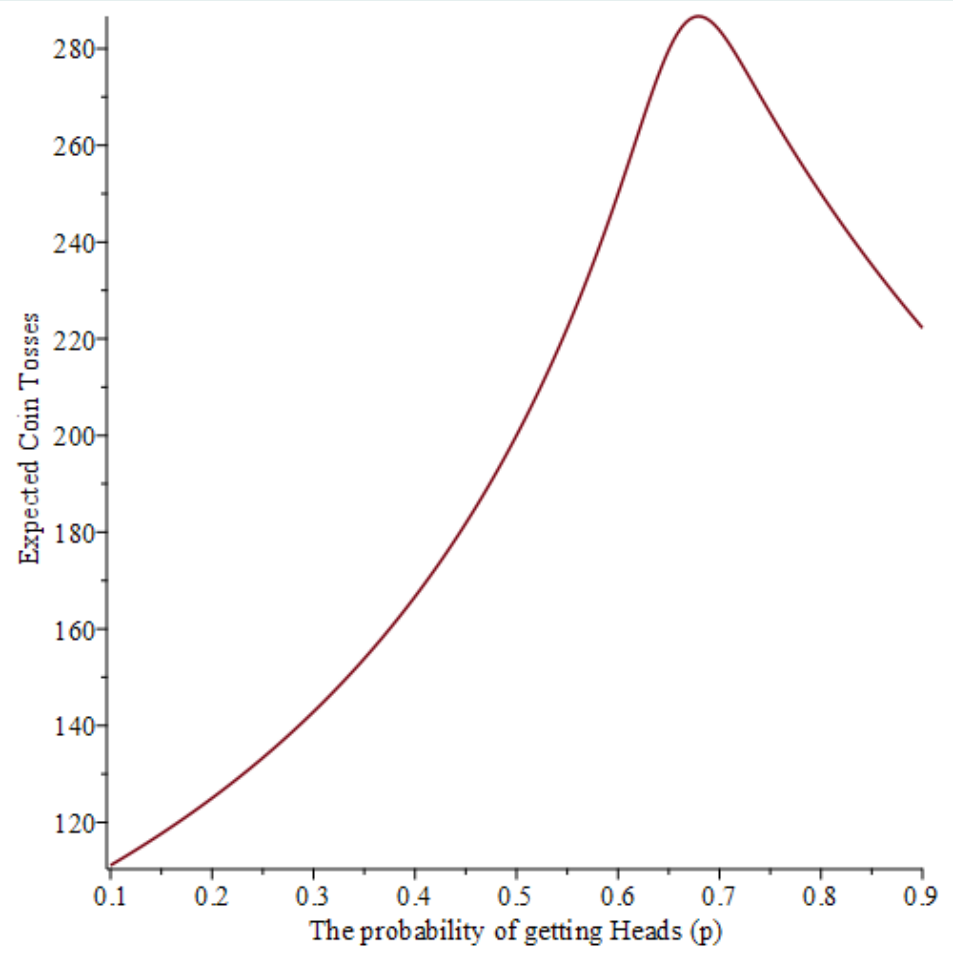
If your goal is to reach either 200 Heads OR 50 Tails,



Expected coin tosses figures for $n = m$

Using the recurrence for the expectation, we can compute the following with various p values.

If your goal is to reach either 200 Heads OR 100 Tails,



Expected coin tosses figures for $n = m$

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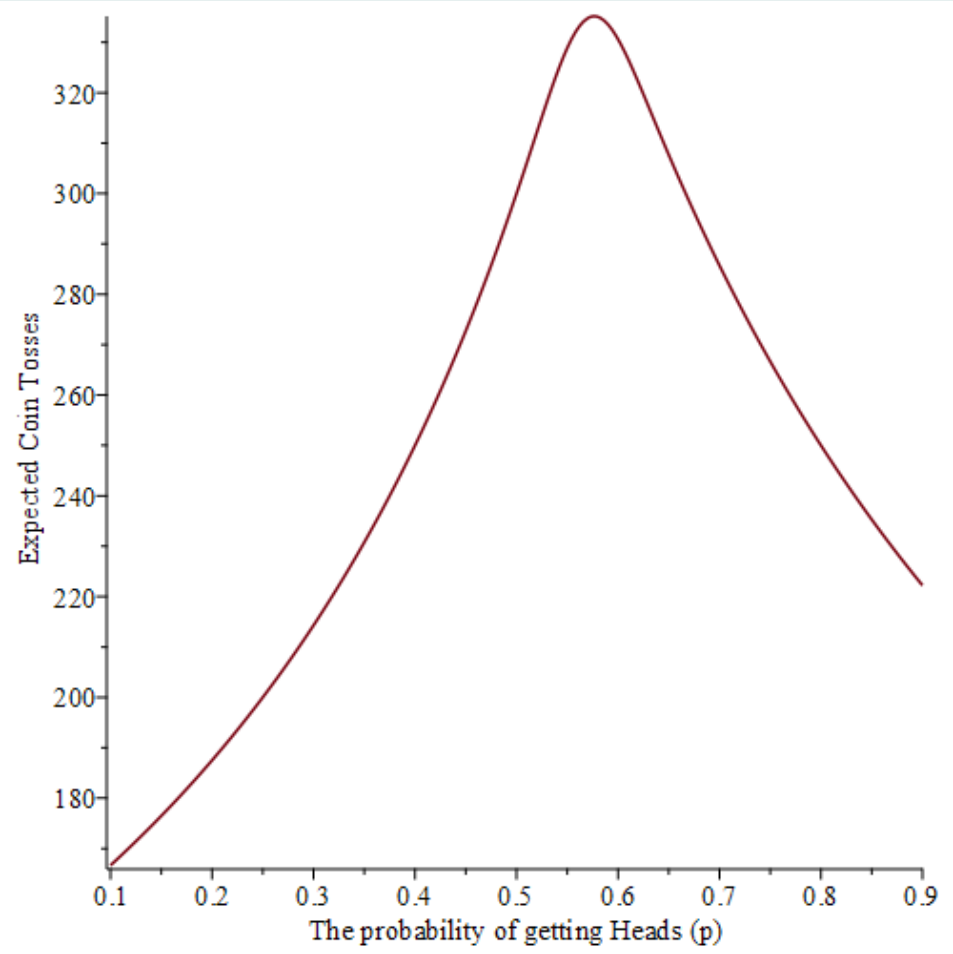
If your goal is to reach either 200 Heads OR 150 Tails,



Expected coin tosses figures for $n = m$

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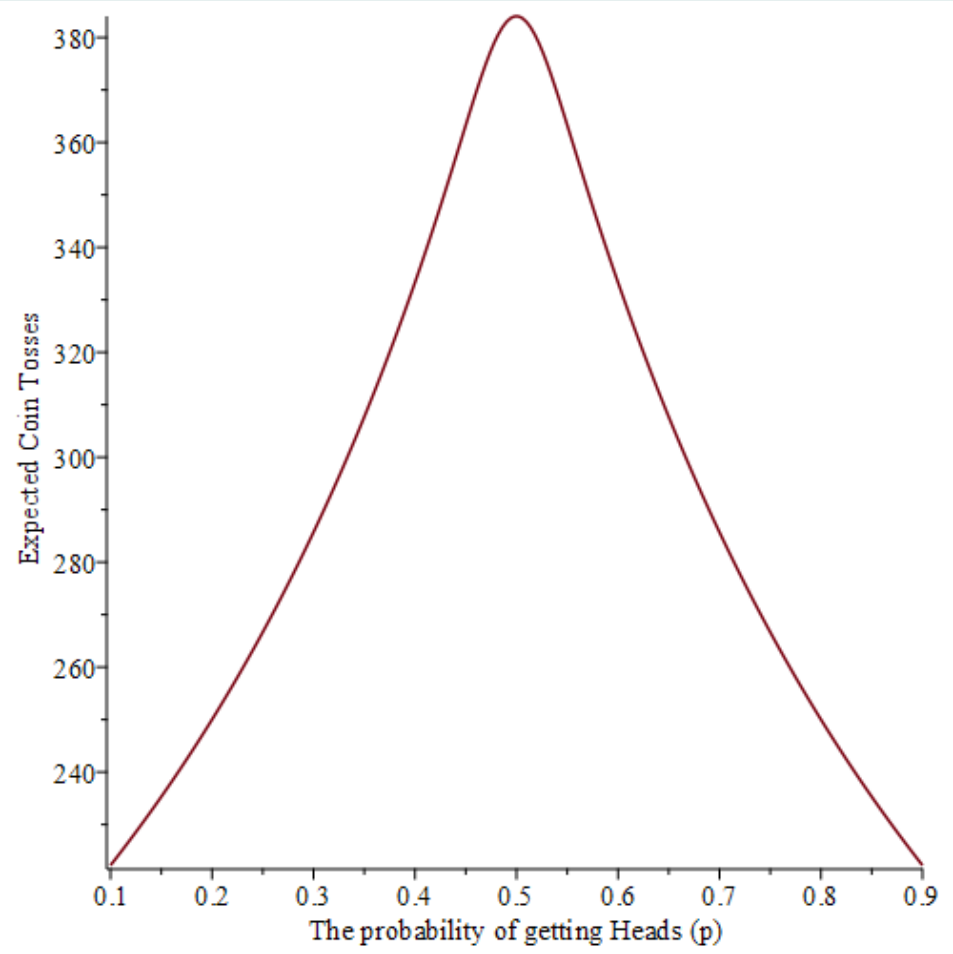

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Remark (2): There is no closed-form expression in terms of r. Once again, we rely on linear recurrences. We use the Zeilberger algorithm!

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$$A(n,1) = 2n - 2B_n, \quad A(n,2) = 4n^2 - 8nB_n, \quad B_n := \frac{n}{4^n}\binom{2n}{n}.$$

Central moments and scaled central moments

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Taking $n \to \infty$, we obtain the same scaled central moments of the **negative binomial distribution!**